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# Zhiming Hu

*Curriculum Vitae*

## Short Bio

Zhiming Hu is a tenure-track Assistant Professor at **The Hong Kong University of Science and Technology (Guangzhou)** starting from August 2025, with a joint appointment at **The Hong Kong University of Science and Technology**. He is a **National Young Talent**. He was a post-doctoral researcher in the **University of Stuttgart**, Germany from August 2022 to July 2025, collaborating with Prof. Andreas Bulling and Prof. Syn Schmitt. He obtained his Ph.D. degree in Computer Software and Theory from **Peking University**, China in 2022, supervised by Prof. Guoping Wang. He received his Bachelor's degree in Optical Engineering from **Beijing Institute of Technology**, China in 2017. His research interests include virtual and augmented reality, human-computer interaction, eye tracking, embodied AI, and human-centered AI. He has published over 30 papers at top venues in VR/AR, HCI, and AI, including SIGGRAPH, TVCG, IEEE VR, ISMAR, CHI, UIST, and AAAI. His work has won **Best Journal Paper Award at ISMAR 2024**, **Best Journal Paper Nominees at IEEE VR 2021**, and **Best Student Paper Nominees at INTERACT 2023**. He is an **Associate Editor of TVCG**. He serves as a reviewer for many top venues, including SIGGRAPH, SIGGRAPH Asia, TVCG, IEEE VR, ISMAR, CHI, UIST, IMWUT, CVPR, ICCV, ECCV, ICLR, AAAI, TMM, IJHCI, and TCSVT.

## Research Interests

His research interests include virtual and augmented reality, human-computer interaction, eye tracking, embodied AI, and human-centered AI. He mainly works on human behavior analysis and modeling for interactive systems with the purpose of understanding human behavior patterns and building human-centered intelligent interactive systems.

## Academic Positions

- **Affiliated Assistant Professor** 2025.10 -
  The Hong Kong University of Science and Technology
- **Assistant Professor** 2025.08 -
  Human-centered Artificial Intelligence Lab
  The Hong Kong University of Science and Technology (Guangzhou)
- **Post-doctoral Researcher** 2022.08 - 2025.07
  Collaborative Artificial Intelligence Lab, Led by Prof. Andreas Bulling

Computational Biophysics and Biorobotics Lab, Led by Prof. Syn Schmitt
University of Stuttgart

## Education

- **Ph.D.** in Computer Software and Theory 2017.09 - 2022.07
  Graphics & Interactive Lab., **Peking University**, Supervised by Prof. Guoping Wang
- **B.Eng.** in Optical Engineering 2013.09 - 2017.07
  School of Optics and Photonics, **Beijing Institute of Technology**

## Awards & Honors

- ISMAR 2026 Recognized Reviewer
- IEEE TVCG 2025 Recognized Reviewer
- **Best Journal Paper Award at ISMAR 2024**
- Baden-Wurttemberg Foundation Postdoctoral Fellowship, 2024
- **Best Student Paper Nominees at INTERACT 2023**
- SimTech Postdoctoral Fellowship, 2022
- National Scholarship (top 2%), 2021
- **Best Journal Paper Nominees at IEEE VR 2021**
- CSC (China Scholarship Council) Scholarship, 2020
- Chancellor's Scholarship (top 2%), 2020
- Leo KoGuan Scholarship (top 5%), 2019
- Leader Scholarship (top 0.2%, 7 out of over 3800 students), 2017
- National Scholarship (top 2%), 2016
- National Scholarship (top 2%), 2014

## Research Projects

- The Coordination Mechanism of Human Eye and Body Movements and Its Correlation Mechanism with Behavioral Semantics, **National Natural Science Foundation of China Youth Project (Category C)** , 2027.01 - 2029.12, 300K RMB, Host
- Study on Mechanisms of Human Visual Attention in Mixed Reality, **National Natural Science Foundation of China General Project**, 2022.01 - 2025.12, 610K RMB, Participate (rank 3/9)
- Machine Learning for Wearable Robotics, **German Research Foundation Excellence Strategy Project**, 2022.08 - 2024.07, 200K EUR, Participate
- Autonomous Robotics iAssistADL, **Baden-Wuerttemberg Research Foundation Project**, 2024.08 - 2025.07, 100K EUR, Participate

## Professional Activities & Talks

**Reviewing**

- Journals: TVCG, IMWUT, TMM, IJHCI, TCSVT, TiiS, MTAP, VR, BRM

- Conferences: SIGGRAPH, SIGGRAPH Asia, CVPR, ICCV, ECCV, CHI, UIST, IEEE VR, ISMAR, ICLR, AAAI, PG, ETRA

**Organizing Committee**

- IEEE AIxVR 2027 Workshop "AIGazeXR: AI-Enhanced Eye Tracking and Gaze Interaction in Extended Reality"
- Executive Committee Member of CCF CAD/CG
- IEEE VR 2027 Program Committee
- AAAI 2027 Program Committee
- ACM MM 2026 Program Committee
- ICXR 2026 Session Chair
- ICXR 2026 Associate Program Co-Chair
- TVCG Associate Editor
- Executive Committee Member of CCF TCVRV
- AAAI 2026 Program Committee
- ETRA 2025 Presentation and Poster Chair
- AAAI 2025 Program Committee
- MuC 2024 Associate Chair
- PETMEI 2024 International Program Committee
- ETRA 2024 Virtualization Chair
- MuC 2023 Associate Chair
- iWOAR 2023 Technical Program Committee

**Invited Talks**

- Investigating the Coordination of Human Eye Gaze and Body Movements in Extended Reality. GAMES Webinar 2025, Hosted by Dr. Xinda Liu, July, 2025.
- Eye-body Coordination during Daily Activities for Gaze Prediction from Full-body Poses. ISMAR 2024, October, 2024.
- Gaze-guided Human Motion Forecasting. IROS 2024 workshop on Nonverbal Cues for Human-Robot Cooperative Intelligence, Hosted by Dr. Jouh Yeong Chew, October, 2024.
- Towards Human-centered Artificial Intelligence. Nanjing University 11th Chengyao Youth Forum, China, December, 2023.
- Towards Human-aware Intelligent User Interfaces. Peking University Fifth Youth Forum on the Next Generation Computer Sciences, China, December, 2023.
- Towards the Coordination of Eye, Body and Context in Daily Activities. Beijing Institute of Technology 10th Teli Forum, China, Hosted by Prof. Guoren Wang, November, 2023.
- The Coordination of Digital Humans. Peking University Career Talk on Computer Science, China, November, 2022.
- Analysis and Prediction of Human Visual Attention in Virtual Reality. Southeast University, China, Hosted by Prof. Ding Ding, June, 2022.
- Recognizing User Tasks from Eye and Head Movements in Immersive Virtual Reality. IEEE VR 2022, Hosted by Prof. Kiyoshi Kiyokawa, March, 2022.

- Forecasting Eye Fixations in Task-Oriented Virtual Environments. GAMES Webinar 2021, Hosted by Prof. Xubo Yang, September, 2021.
- Gaze Analysis and Prediction in Virtual Reality. ChinaVR 2020 - IEEE VR Night, Hosted by Prof. Lili Wang, September 2020.
- Eye-Head Coordination Model for Real-time Gaze Prediction. 2019 International Conference on VR/AR and 3D Display, Hosted by Prof. Feng Xu, June 2019.

## Teaching

- Human-Centered Artificial Intelligence, The Hong Kong University of Science and Technology (Guangzhou), 2026 -
- Machine Perception and Learning, University of Stuttgart, 2022 - 2025

## Publications

* Corresponding author # Equal contribution

***Journal Papers***

1. Chuhan Jiao, **Zhiming Hu***, Andreas Bulling. HAGI++: Head-Assisted Gaze Imputation and Generation. IEEE Transactions on Visualization and Computer Graphics, 2026. (**CCF A**)
2. Takumi Nishiyasu, **Zhiming Hu**, Andreas Bulling, Yoichi Sato. Learning Alignments of Human Gaze and Fine-grained Task Descriptions. Proceedings of the ACM on Computer Graphics and Interactive Techniques, 2026, 9(2): 1-18.
3. Chuhan Jiao, Yao Wang, Guanhua Zhang, Mihai Bâce, **Zhiming Hu**, Andreas Bulling. DiffGaze: A Diffusion Model for Modelling Fine-grained Human Gaze Behaviour on 360° Images. ACM Transactions on Interactive Intelligent Systems, 2026, 16(1): 1-23.
4. **Zhiming Hu***, Guanhua Zhang, Zheming Yin, Daniel Haeufle, Syn Schmitt, Andreas Bulling. HaHeAE: Learning Generalisable Joint Representations of Human Hand and Head Movements in Extended Reality. IEEE Transactions on Visualization and Computer Graphics (oral presentation at ISMAR 2025), 2025, 31(10): 8726 - 8737. (**CCF A**)
5. **Zhiming Hu***, Zheming Yin, Daniel Haeufle, Syn Schmitt, Andreas Bulling. HOIMotion: Forecasting Human Motion During Human-Object Interactions Using Egocentric 3D Object Bounding Boxes. IEEE Transactions on Visualization and Computer Graphics (ISMAR 2024 Journal-track), 2024, 30(11): 7375 - 7385. (**CCF A**, ***Best Journal Paper Award***)
6. **Zhiming Hu***, Jiahui Xu, Syn Schmitt, Andreas Bulling. Pose2Gaze: Eye-body Coordination during Daily Activities for Gaze Prediction from Full-body Poses. IEEE Transactions on Visualization and Computer Graphics (oral presentation at ISMAR 2024), 2025, 31(9): 4655-4666. (**CCF A**)
7. Yao Wang, Yue Jiang, **Zhiming Hu**, Constantin Ruhdorfer, Mihai Bâce, Andreas Bulling. VisRecall++: Analysing and Predicting Visualisation Recallability from Gaze Behaviour. Proceedings of the ACM on Human-Computer Interaction, 2024, 8(ETRA): 1-18.
8. Mayar Elfares, Pascal Reisert, **Zhiming Hu**, Wenwu Tang, Ralf Küsters, Andreas Bulling. Pri-

vatEyes: Appearance-based Gaze Estimation Using Federated Secure Multi-Party Computation. Proceedings of the ACM on Human-Computer Interaction, 2024, 8(ETRA): 1-23.

9. Zehui Lin, Xiang Gu, Sheng Li, **Zhiming Hu**, Guoping Wang. Intentional Head-Motion Assisted Locomotion for Reducing Cybersickness. IEEE Transactions on Visualization and Computer Graphics, 2023, 29(8): 3458-3471. (**CCF A**)
10. **Zhiming Hu**, Andreas Bulling, Sheng Li, Guoping Wang. EHTask: Recognizing User Tasks from Eye and Head Movements in Immersive Virtual Reality. IEEE Transactions on Visualization and Computer Graphics, 2023, 29(4): 1992-2004. (**CCF A**)
11. **Zhiming Hu**, Sheng Li, Meng Gai. Research progress of user task prediction and algorithm analysis (in Chinese). Journal of Graphics, 2021, 42(3): 367-375.
12. **Zhiming Hu**, Andreas Bulling, Sheng Li, Guoping Wang. FixationNet: Forecasting Eye Fixations in Task-Oriented Virtual Environments. IEEE Transactions on Visualization and Computer Graphics (IEEE VR 2021 Journal-track), 2021, 27(5): 2681-2690. (**CCF A**, ***Best Journal Paper Nominees***)
13. **Zhiming Hu**, Sheng Li, Congyi Zhang, Kangrui Yi, Guoping Wang, Dinesh Manocha. DGaze: CNN-Based Gaze Prediction in Dynamic Scenes. IEEE Transactions on Visualization and Computer Graphics (IEEE VR 2020 Journal-track), 2020, 26(5): 1902-1911. (**CCF A**)
14. **Zhiming Hu**, Sheng Li, Meng Gai. Temporal continuity of visual attention for future gaze prediction in immersive virtual reality. Virtual Reality and Intelligent Hardware, 2020, 2(2): 142-152.
15. **Zhiming Hu**, Congyi Zhang, Sheng Li, Guoping Wang, Dinesh Manocha. SGaze: A Data-Driven Eye-Head Coordination Model for Realtime Gaze Prediction. IEEE Transactions on Visualization and Computer Graphics (IEEE VR 2019 Journal-track), 2019, 25(5): 2002-2010. (**CCF A**)

***Conference Papers***

1. Chuhan Jiao, Guanhua Zhang, Yeonjoo Cho, **Zhiming Hu**, Andreas Bulling. DiffEyeSyn: User-specific Subtle Eye Movement Synthesis Using Diffusion Models. Proceedings of the IEEE International Conference on Automatic Face and Gesture Recognition, 2026.
2. Qing Chang#, **Zhiming Hu#***. GazeInterpreter: Parsing Eye Gaze to Generate Eye-Body-Coordinated Narrations. Proceedings of the AAAI Conference on Artificial Intelligence, 2026, 40(21): 17410-17418. (**CCF A**)
3. Chuhan Jiao, **Zhiming Hu***, Andreas Bulling. HAGI: Head-Assisted Gaze Imputation for Mobile Eye Trackers. Proceedings of the ACM Symposium on User Interface Software and Technology, 2025: 1-14. (**CCF A**)
4. **Zhiming Hu***, Daniel Haeufle, Syn Schmitt, Andreas Bulling. HOIGaze: Gaze Estimation During Hand-Object Interactions in Extended Reality Exploiting Eye-Hand-Head Coordination. Proceedings of the ACM Special Interest Group on Computer Graphics and Interactive Techniques, 2025: 1-10. (**CCF A**)
5. Guanhua Zhang, Mohamed Ahmed, **Zhiming Hu***, Andreas Bulling. SummAct: Uncovering

User Intentions Through Interactive Behaviour Summarisation. Proceedings of the ACM CHI Conference on Human Factors in Computing Systems, 2025: 1-17. (**CCF A**)

6. Haodong Yan#, **Zhiming Hu#***, Syn Schmitt, Andreas Bulling. GazeMoDiff: Gaze-guided Diffusion Model for Stochastic Human Motion Prediction. Proceedings of the Pacific Conference on Computer Graphics and Applications, 2024: 1-12.
7. Guanhua Zhang, **Zhiming Hu***, Andreas Bulling. DisMouse: Disentangling Information from Mouse Movement Data via Diffusion Models. Proceedings of the ACM Symposium on User Interface Software and Technology, 2024: 1-13. (**CCF A**)
8. **Zhiming Hu***, Syn Schmitt, Daniel Haeufle, Andreas Bulling. GazeMotion: Gaze-guided Human Motion Forecasting. Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems, 2024: 13017-13022. (**Oral Presentation**)
9. Guanhua Zhang, **Zhiming Hu***, Mihai Bâce, Andreas Bulling. Mouse2Vec: Learning Reusable Semantic Representations of Mouse Behaviour. Proceedings of the ACM CHI Conference on Human Factors in Computing Systems, 2024: 1-17. (**CCF A**)
10. Yao Wang, Weitian Wang, Abdullah Abdelhafez, Mayar Elfares, **Zhiming Hu***, Mihai Bâce, Andreas Bulling. SalChartQA: Question-driven Saliency on Information Visualisations. Proceedings of the ACM CHI Conference on Human Factors in Computing Systems, 2024: 1-14. (**CCF A**)
11. Chuhan Jiao, **Zhiming Hu***, Mihai Bâce, Andreas Bulling. SUPREYES: SUPer Resolution for EYES Using Implicit Neural Representation Learning. Proceedings of the ACM Symposium on User Interface Software and Technology, 2023: 1-13. (**CCF A**)
12. Guanhua Zhang, Matteo Bortoletto, **Zhiming Hu***, Lei Shi, Mihai Bâce, Andreas Bulling. Exploring Natural Language Processing Methods for Interactive Behaviour Modelling. Proceedings of the IFIP Conference on Human-Computer Interaction, 2023: 3-26. (***Best Student Paper Nominees***)

***Short Papers, Abstracts, and Workshops***

1. Duotun Wang, Rui Cao, Kunyu Feng, Wanling Li, Qian Zhong, Zhijing Shao, Yue Jiang, **Zhiming Hu**, Mingming Fan, and Zeyu Wang. Vision-Language Models with In-Context Learning for In-Camera Portrait Composition Guidance. Proceedings of the 2026 ACM Symposium on User Interface Software and Technology Posters, 2026: 1-3.
2. Yi Zou, Ziming Li, Hai-Ning Liang*, **Zhiming Hu***. Contextual Recovery: Guiding Hand Tracking Failures Recovery in Mixed Reality via VLM Reasoning. Proceedings of the 2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops, 2026: 368-372.
3. Mayar Elfares, **Zhiming Hu**, Pascal Reisert, Andreas Bulling, Ralf Küsters. Federated Learning for Appearance-based Gaze Estimation in the Wild. Proceedings of the NeurIPS Workshop Gaze Meets ML, 2023: 20-36.
4. **Zhiming Hu**. Eye Fixation Forecasting in Task-Oriented Virtual Reality. Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops, 2021:

707-708.

5. **Zhiming Hu**. Gaze Analysis and Prediction in Virtual Reality. Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops, 2020: 543-544.